TECHNICAL DILIGENCE REVIEW - RECORD 009

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-009

1. Instruction, scope and the evidence relied on

The subject of the present report is the service architecture, ownership boundaries and dependency structure of the fleet platform, examined for Ardent Peak by Pinehurst Systems Assurance over 3 weeks ending 2025-12-17. The instruction behind the report at TDR-2025-31 was narrow: examine the service architecture, ownership boundaries and dependency structure of the fleet platform and report by 2025-12-17. TDR-2025-31 closes with 8 open items, listed with owner, evidence and the reviewer's view of whether the committed date is realistic. Where the report at TDR-2025-31 cannot conclude, it says what evidence would let it conclude. Pinehurst Systems Assurance raised each finding with the owning engineer before it was written up, and the responses are recorded.

Evidence gathering for TDR-2025-31 was iterative: an initial document request, 19 interviews, then a second request that closed most of the gaps. In report TDR-2025-31 the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. Nothing in TDR-2025-31 rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. L. Casaubon could not verify one class of claim in the present report, namely the behaviour of components deployed only at the two sites not visited. The evidence behind the report at TDR-2025-31 is retained by Pinehurst Systems Assurance and is available to Ardent Peak on request.

L. Casaubon chose the sample for TDR-2025-31 after reading the dependency map, so the selection is not proportionate to headcount or to code volume. Sampling in TDR-2025-31 covered 19 of the 34 catalogued services and 3 of the 18 live sites. L. Casaubon would summarise the present report in one sentence: the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. L. Casaubon's conclusion in the report at TDR-2025-31 is that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. The service architecture, ownership boundaries and dependency structure of the fleet platform is reported here at the level of detail Ardent Peak asked for and no deeper.

Readers of the present report should note that a moderate-evidence finding may carry a high severity, and that this is not a contradiction. The evidence convention in the present report is that severity describes consequence and evidence strength describes confidence, and the two are never merged. Drawing the work together, TDR-2025-31 finds that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. Report TDR-2025-31 concludes that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. The reviewer's tone in TDR-2025-31 is deliberately flat; a finding is not made more serious by the words around it.

Two document requests remained partly unsatisfied when the present report was delivered, and the affected conclusions are marked. L. Casaubon did not visit the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. Interviewees for this report were told that a stated uncertainty would be recorded as an uncertainty rather than resolved against them. 19 interviews inform this report. Where an interview is the sole source for a claim, the claim is graded limited. The company's engineering leadership saw every finding in TDR-2025-31 before delivery and disputed two of them.

Pinehurst Systems Assurance asked the same four questions of every team interviewed for TDR-2025-31, which is how the inconsistencies below were found. Interview evidence in this report came from 19 sessions across 9 teams, each minuted and circulated for correction. Three of the open items in the report at TDR-2025-31 would close on a single document that the company has agreed to produce. L. Casaubon has dated each open item in the present report so that a reader can see which have been open longest rather than only how many there are. The reader of TDR-2025-31 should treat each conclusion as bounded by the section it appears in.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2025-31)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-09-001	Traffic Manager: test report	test report	2025-07-29	Helioscope security	established provenance
EV-09-002	Site Provisioning Service: audit output	audit output	2025-11-28	Helioscope field reliability	verified a stated objective

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-09-003	contract tests, warehouse connectors: vendor statement	vendor statement	2025-05-29	Helioscope field reliability	corroborated an incident narrative
EV-09-004	performance smoke, cloud: release record	release record	2025-12-22	Redwater Security	corroborated an incident narrative
EV-09-005	Speed and Separation Monitor: architecture note	architecture note	2026-08-05	Helioscope security	corroborated an incident narrative
EV-09-006	OTA Updater: incident post-mortem	incident post-mortem	2025-05-08	Helioscope engineering	supported a coverage figure
EV-09-007	E-Stop Chain: design record	design record	2025-05-31	Helioscope security	established provenance
EV-09-008	Safety Controller: configuration export	configuration export	2026-05-15	Meridian Assurance	corroborated an incident narrative
EV-09-009	lint and format: policy document	policy document	2025-05-20	Meridian Assurance	established provenance
EV-09-010	contract tests, warehouse connectors: audit output	audit output	2025-09-18	the sensor vendor	supported a coverage figure
EV-09-011	build, cloud container images: measurement extract	measurement extract	2025-07-29	Helioscope field reliability	supported a coverage figure
EV-09-012	build, cloud container images: design record	design record	2025-11-11	the sensor vendor	established provenance
EV-09-013	hardware-in-the-loop, bench: vendor statement	vendor statement	2026-01-11	the sensor vendor	established a dependency
EV-09-014	build, cloud container images: incident post-mortem	incident post-mortem	2026-02-14	the safety controller vendor	established provenance
EV-09-015	Charge Scheduler: interview minute	interview minute	2026-10-04	the sensor vendor	supported a coverage figure
EV-09-016	Fleet Health Monitor: measurement extract	measurement extract	2026-10-14	the safety controller vendor	evidenced an approval
EV-09-017	Motion Planner: release record	release record	2026-03-09	Redwater Security	evidenced a remediation
EV-09-018	unit tests, robot software: interview minute	interview minute	2025-05-05	Helioscope engineering	established provenance
EV-09-019	Robot Shadow Store: vendor statement	vendor statement	2026-10-28	Helioscope engineering	corroborated an incident narrative
EV-09-020	build, cloud container images: architecture note	architecture note	2025-05-21	Helioscope security	verified a stated objective
EV-09-021	Safety Controller: vendor statement	vendor statement	2026-07-23	Helioscope field reliability	corroborated an incident narrative
EV-09-022	Safety Controller: design record	design record	2026-03-14	Meridian Assurance	evidenced a remediation
EV-09-023	Payload Scale Interface: design record	design record	2026-08-13	Helioscope safety	evidenced a remediation
EV-09-024	WMS Connector: audit output	audit output	2025-11-01	Helioscope safety	evidenced a remediation
EV-09-025	Simulation Harness: policy document	policy document	2025-12-30	Helioscope safety	corroborated an incident narrative
EV-09-026	dependency and licence scan: interview minute	interview minute	2025-05-19	Helioscope engineering	established a dependency
EV-09-027	integration tests, cloud: audit output	audit output	2026-05-09	Helioscope engineering	evidenced an approval
EV-09-028	Billing Usage Meter: vendor statement	vendor statement	2025-05-18	Meridian Assurance	evidenced an approval
EV-09-029	Robot Shadow Store: vendor statement	vendor statement	2025-08-20	Helioscope safety	supported a coverage figure
EV-09-030	Costmap Fusion: interview minute	interview minute	2026-02-10	the safety controller vendor	supported a capacity conclusion
EV-09-031	Analytics Warehouse	configuration export	2025-05-30	Meridian Assurance	established a

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
	Loader: configuration export				dependency
EV-09-032	unit tests, robot software: policy document	policy document	2025-12-16	the safety controller vendor	verified a stated objective
EV-09-033	Hardware-in-the-Loop Rig: test report	test report	2025-09-20	the sensor vendor	evidenced an approval
EV-09-034	E-Stop Chain: architecture note	architecture note	2025-09-25	the sensor vendor	verified a stated objective
EV-09-035	Billing Usage Meter: configuration export	configuration export	2026-03-20	Meridian Assurance	evidenced an approval
EV-09-036	build, robot firmware images: policy document	policy document	2026-08-28	Redwater Security	established provenance

1. Instruction, scope and the evidence relied on - part 2 of 2

Below, the 1. Instruction, scope and the evidence relied on continues; nothing separates this part from the last except the page on which it happens to fall. Pagination alone determines where it falls.

Because the basis of the 1. Instruction, scope and the evidence relied on is unchanged, figures in this part may be compared directly with those already given. The parts are to be read together.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2025-31) (continued, part 2 of 2)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-09-037	simulation suite, short: audit output	audit output	2025-09-20	Helioscope field reliability	verified a stated objective
EV-09-038	unit tests, robot software: measurement extract	measurement extract	2025-10-20	the sensor vendor	corroborated an incident narrative
EV-09-039	Traffic Manager: measurement extract	measurement extract	2025-07-22	Meridian Assurance	evidenced an approval
EV-09-040	Release Orchestrator: policy document	policy document	2026-06-09	Meridian Assurance	evidenced an approval
EV-09-041	Config Distribution: release record	release record	2025-10-06	Meridian Assurance	supported a capacity conclusion
EV-09-042	lint and format: design record	design record	2025-12-21	Redwater Security	supported a coverage figure
EV-09-043	unit tests, robot software: incident post-mortem	incident post-mortem	2026-09-29	Helioscope field reliability	evidenced an approval
EV-09-044	Gripper Controller: configuration export	configuration export	2026-09-22	the safety controller vendor	established a dependency
EV-09-045	Lidar Driver: interview minute	interview minute	2025-07-22	the sensor vendor	evidenced an approval
EV-09-046	container image vulnerability scan: audit output	audit output	2025-07-14	Meridian Assurance	supported a capacity conclusion
EV-09-047	Wheel Odometry Filter: audit output	audit output	2025-11-01	Helioscope field reliability	evidenced an approval
EV-09-048	lint and format: design record	design record	2026-01-13	the sensor vendor	verified a stated objective
EV-09-049	Analytics Warehouse Loader: release record	release record	2026-02-17	Redwater Security	established a dependency
EV-09-050	hardware-in-the-loop, bench: test report	test report	2026-08-27	the sensor vendor	evidenced an approval
EV-09-051	Payload Scale Interface: interview minute	interview minute	2025-09-03	Helioscope security	corroborated an incident narrative
EV-09-052	Traffic Manager: architecture note	architecture note	2026-08-15	Helioscope engineering	verified a stated objective
EV-09-053	Order Broker: test report	test report	2026-08-22	Meridian Assurance	supported a coverage figure
EV-09-054	E-Stop Chain: architecture note	architecture note	2025-06-24	Helioscope security	supported a capacity conclusion
EV-09-055	build, cloud container	policy document	2026-03-30	Meridian Assurance	evidenced a

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
	images: policy document				remediation
EV-09-056	Wheel Odometry Filter: incident post-mortem	incident post-mortem	2026-05-08	Meridian Assurance	evidenced an approval
EV-09-057	build, robot firmware images: measurement extract	measurement extract	2026-02-18	Meridian Assurance	verified a stated objective
EV-09-058	integration tests, cloud: incident post-mortem	incident post-mortem	2026-03-13	Helioscope engineering	evidenced a remediation
EV-09-059	OTA Updater: configuration export	configuration export	2025-06-30	Helioscope engineering	evidenced a remediation
EV-09-060	Speed and Separation Monitor: audit output	audit output	2025-06-17	Redwater Security	verified a stated objective
EV-09-061	Wheel Odometry Filter: release record	release record	2025-10-14	Helioscope field reliability	established provenance
EV-09-062	performance smoke, cloud: audit output	audit output	2026-06-11	Helioscope field reliability	supported a capacity conclusion
EV-09-063	Order Broker: release record	release record	2025-09-09	Redwater Security	established a dependency
EV-09-064	static analysis, robot software: interview minute	interview minute	2025-10-18	the sensor vendor	verified a stated objective
EV-09-065	dependency and licence scan: release record	release record	2025-07-25	Redwater Security	evidenced a remediation
EV-09-066	Speed and Separation Monitor: measurement extract	measurement extract	2026-08-14	Helioscope engineering	supported a capacity conclusion
EV-09-067	Robot Shadow Store: audit output	audit output	2025-06-28	Meridian Assurance	established provenance
EV-09-068	static analysis, robot software: test report	test report	2025-08-09	Meridian Assurance	established provenance
EV-09-069	Charge Scheduler: policy document	policy document	2026-06-18	Helioscope security	evidenced a remediation
EV-09-070	Battery Management Service: audit output	audit output	2026-06-25	the sensor vendor	supported a coverage figure
EV-09-071	Motion Planner: vendor statement	vendor statement	2025-12-10	Redwater Security	established a dependency
EV-09-072	Charge Scheduler: architecture note	architecture note	2026-09-04	Helioscope field reliability	established a dependency

2. Video Snippet Service and the services on the control path

Video Snippet Service's dependency ageing was checked: a transcoder build with known but low-severity upstream advisories. The assessment of Video Snippet Service is not unqualified: the reviewer accepts the engineering position while noting that the verifying test does not exist. Applications publishes one objective for Video Snippet Service: a requested snippet is available to the operator inside 45 s. Applications tracks 12 open defects on Video Snippet Service, including the acknowledged case where the retention lifecycle was applied per bucket, so one region kept clips for ninety days. Underlying Video Snippet Service in this report are the retention configuration, a lifecycle audit across regions and the access log sample. Video Snippet Service rests on a single source, and the reviewer has said so rather than implying corroboration.

Video Snippet Service shows 43 per cent line coverage in the pipeline, and the reviewer regards that as adequate for the risk this component carries. The Video Snippet Service codebase is Go 1.22 with a media pipeline throughout, deployed onto cloud, containers plus a media transcoder, and its durable record is object storage with a thirty-day lifecycle and a PostgreSQL 14 index. Video Snippet Service is promoted monthly, with the ring order fixed by policy. Deployment of Video Snippet Service is cloud, regional, with an on-robot capture buffer, which sets the blast radius of any defect in it. The dependency map places the on-robot camera buffer and the identity service above Video Snippet Service and the remote operations console and the incident review workflow below it. Video Snippet Service would be a candidate for the first post-close engineering review under any reasonable plan.

Alarm Router was built in Python 3.11 and has not been rewritten; it still runs on cloud, containers plus a scheduled dispatcher against PostgreSQL 14 for routes and delivery receipts. Service-level intent for Alarm Router states that a dispatched alarm reaches an on-call responder inside 90 s at p95. On Alarm Router the conclusion is qualified: the component is adequate today and carries a supportability liability that is not on the funded plan. The topology for Alarm Router is cloud, regional, with a second delivery provider, and the reviewer confirmed it against the deployment manifests. Alarm Router sits at 51 per cent line coverage; the trend across the reviewed period was up by three points. Alarm Router was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Alarm Router is owned by the Field Reliability team, with Wanda Njoku named as the engineering owner of record. The paging provider and the customer notification channel rely on Alarm Router, which in turn relies on the fleet health monitor and the on-call rotation source. Alarm Router releases fortnightly. For Alarm Router the reviewer worked from the routing table export, delivery receipts for one quarter and two missed-page reviews. Field Reliability tracks 14 open defects on Alarm Router, including the acknowledged case where an on-call rotation gap in the routing table falls back to a shared mailbox nobody watches at night. Alarm Router is reported at the evidence strength available and has not been promoted to fit the narrative.

The Identity and Authorization codebase is Go 1.22 throughout, deployed onto cloud, containers on a managed Kubernetes cluster, and its durable record is PostgreSQL 14 with a hardware-backed key store. The evidence base for Identity and Authorization was the authorization model document, a token lifetime audit and the secret rotation log. Identity and Authorization's dependency ageing was checked: none identified in the reviewed manifests. Deployment of Identity and Authorization is cloud, global control plane with regional caches, which sets the blast radius of any defect in it. Every cloud service and the remote operations console rely on Identity and Authorization, which in turn relies on the customer directory federations and the secret store. Identity and Authorization is the clearest instance in TDR-2025-31 of engineering practice being better than the documentation suggests.

Identity and Authorization commits that token issuance inside 200 ms at p95 and no unauthenticated write path, and the objective is measured continuously rather than sampled. Identity and Authorization was checked for shared-database coupling, and the ownership boundary held. Identity and Authorization shows 84 per cent line coverage in the pipeline, and coverage alone overstates the assurance here, because the bench suite is the real safety net. The Security Engineering team carries Identity and Authorization on its service inventory, and Sonia Krall answered for it in interview. Identity and Authorization was weighed on its coverage, its defect backlog and its dependency position, and it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Identity and Authorization was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Against Traffic Manager the backlog runs to 12 items, and it includes a known behaviour whereby a corridor reservation leaks when a robot drops off the network mid-traverse. Traffic Manager commits that corridor deadlock detected and broken inside two seconds, and the objective is measured continuously rather than sampled. The manifests place Traffic Manager cloud, regional, with a degraded on-robot yield behaviour if the link drops. Evidence examined for Traffic Manager comprised the reservation protocol specification and a replay of three deadlock events. Inputs to Traffic Manager come from the fleet allocator and the site map service, and its consumers are the motion planner and the remote operations console. Traffic Manager is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Traffic Manager reports 74 per cent line coverage, measured by the same instrumentation the pipeline uses. Read against its objective, Traffic Manager there is no finding of unfitness; there is a finding that the team cannot currently demonstrate fitness. Traffic Manager does not share a schema with any consumer; every read by the motion planner and the remote operations console goes through the published interface. Owen Brantley leads Fleet Services's work on Traffic Manager and was the reviewer's principal interview for it. Traffic Manager pairs a Go 1.22 implementation with cloud, containers on a managed Kubernetes cluster, and its data lives in Redis 7 for corridor reservations with a PostgreSQL 14 audit trail. Traffic Manager has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Schedule A - Service inventory, ownership and objective attainment (TDR-2025-31)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Motion Planner	Rust 1.79	Autonomy Core	71%	plan cycle completes inside 40 ms at the 99th percentile	the costmap fusion pipeline and the localization engine	none; the toolchain is pinned to a supported compiler release
Costmap Fusion	C++17	Perception	58%	fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms	the lidar driver, the depth camera bridge and the wheel odometry filter	a vendored geometry library whose upstream branch left support in 2024
Fleet Allocator	Go 1.22	Fleet Services	82%	task assignment inside 900 ms of order arrival at p95	the order broker and the robot shadow store	none identified in the reviewed manifests
Traffic Manager	Go 1.22	Fleet Services	74%	corridor deadlock detected and broken inside two seconds	the fleet allocator and the site map service	Redis 6 client library still vendored in one code path
Charge Scheduler	Python 3.11	Fleet Services	63%	fleet state of charge stays above 35 per cent through the peak shift at every site	the battery management service and the site shift calendar	none; the dependency set was refreshed in the 2026 platform upgrade
Safety Controller	C, written to a constrained coding subset	Safety Engineering	91%	a validated protective-stop trigger de-energises the drive inside 120 ms	the E-stop chain, the speed and separation monitor and the drive current sensors	none; the supplier's support commitment runs to 2031
E-Stop Chain	VHDL	Safety Engineering	witnessed test	drive power is removed within 250 ms of button actuation on every tested unit	the operator stop buttons and the perimeter interlocks	none; the relay parts are current production
Lidar Driver	C++17	Robot Firmware	44%	no more than one dropped scan per ten thousand at nominal spin rate	the sensor hardware and the time synchronisation daemon	the vendor's software development kit is two major versions behind current
Localization Engine	C++17	Autonomy Core	66%	pose error stays under 25 mm in mapped aisles at survey speed	the lidar driver, the wheel odometry filter and the map distribution channel	an unmaintained scan-matching library is still linked in the fallback path
WMS Connector	Java 21	Integrations	69%	order acknowledgement returned to the warehouse system inside 1.5 s at p95	the customer warehouse management systems and the identity service	one customer integration still runs on a connector build from 2024
Order Broker	Go 1.22	Fleet Services	78%	no accepted order is lost, measured by journal reconciliation each night	the WMS connector and the site configuration service	none identified
Telemetry Ingest	Go 1.22	Data and Telemetry	61%	telemetry visible in the operator console inside 30 s of emission at p95	the edge gateway agent and the event bus	the stream processor runs a release that leaves community support in late 2026
OTA Updater	Rust 1.79 on the robot, Go 1.22 in the cloud	Platform Infrastructure	76%	an update is either fully applied or fully rolled back, with no mixed-version robot	the release orchestrator and the artefact signing service	none; artefact signing was moved to the current key format in 2025
Identity and	Go 1.22	Security	84%	token issuance	the customer	none identified in

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Authorization		Engineering		inside 200 ms at p95 and no unauthenticated write path	directory federations and the secret store	the reviewed manifests
Remote Operations Console	TypeScript	Applications	52%	operator input reaches the robot inside 350 ms at p95 on a supported network	the identity service, the telemetry ingest path and the video snippet service	a charting library that no longer receives security updates
Simulation Harness	Python 3.11	Developer Productivity	47%	the nightly scenario suite completes inside four hours	the planner and localization builds under test	the physics core depends on a library branch that stopped receiving fixes in 2023
Hardware-in-the-Loop Rig	Python 3.11 test harness around production firmware	Robot Firmware	39%	every firmware candidate runs the full bench suite before ring one	the firmware candidate artefacts and the bench sensor stimulators	the rig controller runs an operating system release that left vendor support in 2025

2. Video Snippet Service and the services on the control path - part 2 of 2

Because the 2. Video Snippet Service and the services on the control path does not fit one page, its remainder is set out below under the same references. Pagination alone determines where it falls.

Readers should carry the 2. Video Snippet Service and the services on the control path's stated conventions across the break, since this part adds no definition of its own. The break is editorial.

Schedule A - Service inventory, ownership and objective attainment (TDR-2025-31) (continued, part 2 of 2)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Fleet Health Monitor	Python 3.11	Field Reliability	57%	a fleet-affecting condition raises an alert inside three minutes of first symptom	the telemetry ingest path and the robot shadow store	none identified
Battery Management Service	C on the pack controller, Python 3.11 in the cloud	Robot Firmware	54%	pack cell imbalance is reported before it reaches the derating threshold	the pack sensors and the charge scheduler	one pack revision uses a cell supplier that has since discontinued the part
Gripper Controller	C++17	Robot Firmware	41%	a grasp completes or reports failure inside 1.2 s	the payload scale interface and the perception grasp proposals	none; the coprocessor part is current production
Map Authoring Studio	TypeScript	Applications	49%	a published map revision reaches every robot at the site inside fifteen minutes	the site survey imports and the identity service	a geometry rendering library one major version behind its supported line
Site Provisioning Service	Go 1.22	Platform Infrastructure	64%	a new site reaches a working configuration inside one working day	the identity service and the customer network intake records	none identified
Event Bus	operated open-source broker, no in-house code	Platform Infrastructure	witnessed test	publish acknowledgement inside 25 ms at p95 and no partition without a leader	every publishing service	the broker minor release in production leaves the vendor's support window in 2027
Config Distribution	Rust 1.79 agent	Platform Infrastructure	72%	a configuration change reaches every online robot inside five minutes	the site provisioning service and the artefact signing service	none; the bundle format was revised in 2025
Robot Shadow	Go 1.22	Fleet Services	68%	shadow state is no	the telemetry	none identified

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Store				more than two seconds behind the robot at p95	ingest path and the edge gateway agent	
Analytics Warehouse Loader	Python 3.11	Data and Telemetry	45%	the daily load completes before 06:00 local time at every region	the telemetry ingest path, the order broker and the billing usage meter	one transformation still runs on a deprecated warehouse function
Billing Usage Meter	Go 1.22	Data and Telemetry	79%	metered usage reconciles to the order journal within 0.5 per cent each month	the order broker and the WMS connector	none identified
Alarm Router	Python 3.11	Field Reliability	51%	a dispatched alarm reaches an on-call responder inside 90 s at p95	the fleet health monitor and the on-call rotation source	the paging provider's older application interface is still in use
Video Snippet Service	Go 1.22	Applications	43%	a requested snippet is available to the operator inside 45 s	the on-robot camera buffer and the identity service	a transcoder build with known but low-severity upstream advisories
Speed and Separation Monitor	C++17	Safety Engineering	87%	commanded speed is reduced before the separation distance falls below the protective distance	the costmap fusion pipeline and the safety-rated scanner inputs	none; the sensing hardware is current production
Wheel Odometry Filter	C++17	Robot Firmware	62%	odometry drift stays under 1.5 per cent of distance travelled on a clean floor	the wheel encoders and the inertial measurement unit	none identified
Payload Scale Interface	C++17	Robot Firmware	36%	payload mass is reported within 2 per cent of the reference load	the load cell hardware	none identified
Edge Gateway Agent	Rust 1.79	Platform Infrastructure	74%	no telemetry is lost for an outage shorter than four hours	the on-robot services that emit telemetry	none; the transport library was updated in early 2026
Release Orchestrator	Go 1.22	Developer Productivity	55%	no artefact reaches production without a recorded approval and a passing gate set	the build system, the simulation harness gate and the identity service	none identified

3. Deployment topology and failure domains, beginning with Fleet Health Monitor

Fleet Health Monitor owns its own data and exposes it only through a versioned interface, which the reviewer read. Fleet Health Monitor's dependency ageing was checked: none identified. Responsibility for Fleet Health Monitor was confirmed with Wanda Njoku, whose team, Field Reliability, also holds the on-call duty. What the reviewer had for Fleet Health Monitor was the alert rule inventory, three months of case records and a rule coverage review; the conclusions below do not reach past it. The manifests place Fleet Health Monitor cloud, regional. Fleet Health Monitor was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Automated test coverage on Fleet Health Monitor stands at 57 per cent line coverage. The published objective for Fleet Health Monitor is that a fleet-affecting condition raises an alert inside three minutes of first symptom. On Fleet Health Monitor the conclusion is qualified: the reviewer could corroborate the availability claim but not the latency claim, which rests on a single dashboard. Changes to Fleet Health Monitor reach production fortnightly. 17 defects remain open

against Fleet Health Monitor; the one the reviewer pressed on is that alert grouping collapses distinct faults at the same site into one case, hiding the second fault. Fleet Health Monitor is reported at the evidence strength available and has not been promoted to fit the narrative.

The Costmap Fusion codebase is C++17 throughout, deployed onto on-robot Linux, pinned to the drive control core, and its durable record is a shared-memory ring buffer with no durable persistence. Daniel Okafor of Perception maintains Costmap Fusion, and the service inventory entry was current at the review date. Evidence examined for Costmap Fusion comprised the fusion design note, sensor timing captures and the reviewer's own bench replay. Costmap Fusion was tested against the dependency ageing register, which records a vendored geometry library whose upstream branch left support in 2024. For Costmap Fusion the reviewer records that it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Costmap Fusion was cross-checked against the engineering record supplied for TDR-2025-31, and the two agree.

Costmap Fusion sits at 58 per cent line coverage; the trend across the reviewed period was down by two points after a large refactor. Interface discipline around Costmap Fusion is adequate; the schema is versioned and change is gated on a compatibility check. 21 items sit in the Costmap Fusion backlog; the entry that bears on the investment case is that a stale lidar frame is retained for one extra cycle while a driver reconnects. Costmap Fusion releases monthly, batched with perception model updates. Inputs to Costmap Fusion come from the lidar driver, the depth camera bridge and the wheel odometry filter, and its consumers are the motion planner and the speed and separation monitor. Costmap Fusion was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Data ownership is clear for Gripper Controller: it writes a local parameter store and no other service writes there. Reading the dependency map from Gripper Controller outward gives the payload scale interface and the perception grasp proposals on the input side and the order execution path and the safety controller fault input on the output side. Gripper Controller ships quarterly; the reviewer sampled the last twelve releases. The assessment of Gripper Controller is not unqualified: the component would not block the transaction, and it would appear on any competent post-close engineering plan. Gripper Controller rests on the gripper control specification, calibration records for eleven units and a force bench test so far as this review is concerned. Gripper Controller was raised again with the owning engineer after the initial interview, and the position did not change.

The open defect backlog for Gripper Controller holds 15 items, the most consequential being that force limiting relies on a calibration that is only re-run when a gripper is replaced. Placement of Gripper Controller is edge, only on the two robot variants fitted with a manipulator, which the reviewer tested by tracing a request end to end. Robot Firmware holds Gripper Controller, and the ownership entry naming Ines Duarte matches the on-call rotation. C++17 is the implementation language for Gripper Controller, which is hosted on on-robot Linux with a firmware coprocessor with a local parameter store behind it. Where Gripper Controller is concerned, the commitment is that a grasp completes or reports failure inside 1.2 s; the reviewer traced the metric to its source. Gripper Controller is the reason one section of TDR-2025-31 concludes narrowly rather than generally.

One coupling concern applies to Payload Scale Interface: a local calibration record is shared with a second service, so a schema change is not a local decision. On paper Payload Scale Interface releases as needed, and the observed practice did not diverge from that. Against Payload Scale Interface, the coverage figure is 36 per cent line coverage, which the reviewer notes that the untested share includes the error handling. The published objective for Payload Scale Interface is that payload mass is reported within 2 per cent of the reference load. Payload Scale Interface has a single named owner, Ines Duarte of Robot Firmware, which is not true of every service reviewed. Payload Scale Interface was tested rather than accepted, and the test is described in the working papers behind TDR-2025-31.

Robot Firmware tracks 11 open defects on Payload Scale Interface, including the acknowledged case where the calibration record is stored locally and is lost when the compute module is swapped. Payload Scale Interface was built in C++17 and has not been rewritten; it still runs on on-robot Linux, user-space driver against a local calibration record. Payload Scale Interface sits edge, only on robots fitted with the weighing deck, and no alternative placement is currently supported. The reviewer's judgement on Payload Scale Interface is that it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Underlying Payload Scale Interface in this report are the interface specification, calibration records for nine units and a module-swap test. Payload Scale Interface was tested rather than accepted, and the test is described in the working papers behind TDR-2025-31.

For Fleet Allocator the supportability position is none identified in the reviewed manifests. Deployment of Fleet Allocator is cloud, one regional deployment serving all sites in that region, which sets the blast radius of any defect in it. Data ownership is clear for Fleet Allocator: it writes PostgreSQL 14 with a synchronous standby and no other service writes there. The evidence base for Fleet Allocator was the allocator service contract, two design reviews and a week of assignment audit logs. Fleet Services publishes one objective for Fleet Allocator: task assignment inside 900 ms of order arrival at p95. Fleet Allocator was one of two items where TDR-2025-31 could not obtain the underlying data within the engagement.

Fleet Allocator sits between the order broker and the robot shadow store and the traffic manager, the charge scheduler and the fleet health monitor, and that position is what makes its availability objective load-bearing. Fleet Allocator sits at 82 per cent line coverage; the trend across the reviewed period was down by two points after a large refactor. The assessment of Fleet Allocator is not unqualified: the component would not block the transaction, and it would appear on any competent post-close engineering plan. Marta Kellerman leads Fleet Services's work on Fleet Allocator and was the reviewer's principal interview for it. Fleet Allocator pairs a Go 1.22 implementation with cloud, containers on a managed Kubernetes cluster, and its data lives in PostgreSQL 14 with a synchronous standby. Fleet Allocator is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Schedule B - Deployment topology and failure domain by service (TDR-2025-31)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Motion Planner	edge, one instance per robot with no cloud fallback	every three weeks, promoted ring by ring	5	one robot	Priya Raghunathan
Costmap Fusion	edge, co-scheduled with the planner on the compute module	monthly, batched with perception model updates	6	one robot	Daniel Okafor
Fleet Allocator	cloud, one regional deployment serving all sites in that region	weekly, behind a change approval	5	one region	Marta Kellerman
Traffic Manager	cloud, regional, with a degraded on-robot yield behaviour if the link drops	weekly, paired with the allocator release	3	one region	Owen Brantley
Charge Scheduler	cloud, regional, with a conservative local fallback policy on the robot	fortnightly, without a formal approval gate until March 2026	2	one region	Nadia Semenova
Safety Controller	edge, dual-channel, with cross-checking between channels	twice a year, with full revalidation of the safety case	1	one robot	Bettina Vogel
E-Stop Chain	edge, hardwired, independent of the compute module	changed only by an engineering change order	4	one robot	Theo Bassett
Lidar Driver	edge, one process per sensor	as needed, historically twice a year	2	one robot	Yuki Harada
Localization Engine	edge, with map distribution from the cloud	every three weeks with the planner	2	one robot	Rahul Menon
WMS Connector	cloud, one deployment per customer tenant	every two weeks, per tenant on request	2	one region	Tomas Villareal
Order Broker	cloud, regional	weekly	6	one region	Grace Lindqvist
Telemetry Ingest	cloud, one ingest endpoint per region with an edge buffer on the robot	weekly, with schema changes gated on a compatibility check	3	one robot	Farida Cisse
OTA Updater	split edge and cloud, with staged rings and a per-site pause control	every three weeks, following the autonomy release	3	one robot	Erik Palmgren
Identity and Authorization	cloud, global control plane with regional caches	weekly, with a security review on every change	2	one region	Sonia Krall
Remote Operations Console	cloud, with a thin session relay in each	twice a week	2	one region	Anjali Sekhon

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
	region				
Simulation Harness	cloud batch, with a local mode for developers	continuous, from the main branch	1	one region	Colin Achebe
Hardware-in-the-Loop Rig	laboratory, two rigs, one of which lags the current sensor configuration	the harness changes with each firmware release	3	one region	Nathan Oduya

3. Deployment topology and failure domains, beginning with Fleet Health Monitor - part 2 of 2

A further extract of the 3. Deployment topology and failure domains, beginning with Fleet Health Monitor appears here, taken in order from the point at which the previous part ended. The parts are to be read together.

Scope, cut-off and units for the 3. Deployment topology and failure domains, beginning with Fleet Health Monitor were fixed at its first part and are not repeated, being unaltered. No row is omitted at the join.

Schedule B - Deployment topology and failure domain by service (TDR-2025-31) (continued, part 2 of 2)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Fleet Health Monitor	cloud, regional	fortnightly	3	one region	Wanda Njoku
Battery Management Service	split edge and cloud	pack firmware twice a year, the cloud service monthly	3	one robot	Peter Halvorsen
Gripper Controller	edge, only on the two robot variants fitted with a manipulator	quarterly	3	one robot	Ines Duarte
Map Authoring Studio	cloud, single global deployment	weekly	6	one region	Lucia Ferrante
Site Provisioning Service	cloud, global	fortnightly	6	one region	Marcus Duplessis
Event Bus	cloud, one cluster per region, three availability zones	upgraded twice a year	6	one region	Erik Palmgren
Config Distribution	split edge and cloud, with a signed bundle format	weekly	5	one robot	Sam Oyelaran
Robot Shadow Store	cloud, regional, sharded by site	weekly	2	one region	Marta Kellerman
Analytics Warehouse Loader	cloud, one loader per region writing to a shared warehouse	weekly	4	one region	Farida Cisse
Billing Usage Meter	cloud, regional	monthly, aligned to the billing cycle	5	one region	Nadia Semenova
Alarm Router	cloud, regional, with a second delivery provider	fortnightly	1	one region	Wanda Njoku
Video Snippet Service	cloud, regional, with an on-robot capture buffer	monthly	3	one region	Anjali Sekhon
Speed and Separation Monitor	edge, cross-checked against the safety controller's own inputs	twice a year, with revalidation	1	one robot	Bettina Vogel
Wheel Odometry Filter	edge	with the firmware, twice a year	2	one robot	Yuki Harada
Payload Scale Interface	edge, only on robots fitted with the weighing deck	as needed	1	one robot	Ines Duarte
Edge Gateway Agent	edge, one per robot, with mutual authentication to the cloud	every three weeks	5	one robot	Sam Oyelaran
Release Orchestrator	cloud, single global deployment	continuous, self-hosting	5	one region	Colin Achebe

4. Interface boundaries and data ownership around Battery Management Service

The evidence base for Battery Management Service was the pack firmware release notes, cell test data for two revisions and a derating incident file. One coupling concern applies to Battery Management Service: pack non-volatile memory and a cloud time-series store is shared with a second service, so a schema change is not a local decision. Ownership of Battery Management Service sits with Robot Firmware; Peter Halvorsen signs off changes to it. Where Battery Management Service is concerned, the commitment is that pack cell imbalance is reported before it reaches the derating threshold; the reviewer traced the metric to its source. Battery Management Service was assessed against the objective and the defect record, and it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Battery Management Service was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The pack sensors and the charge scheduler feed Battery Management Service; a failure in Battery Management Service is felt by the charge scheduler and the fleet health monitor. Battery Management Service sits at 54 per cent line coverage; the trend across the reviewed period was up by four points. Pack non-volatile memory and a cloud time-series store is the system of record for Battery Management Service, a C on the pack controller, Python 3.11 in the cloud component running on split: pack firmware and a cloud analytics service. Deployment of Battery Management Service is split edge and cloud, which sets the blast radius of any defect in it. Battery Management Service ships pack firmware twice a year, the cloud service monthly; the reviewer sampled the last twelve releases. Battery Management Service would move from a concern to a verified defect if one further measurement were supplied.

Platform Infrastructure releases Config Distribution weekly, and no release in the sampled window bypassed the gate set. Rust 1.79 agent with a Go 1.22 control service is the implementation language for Config Distribution, which is hosted on split: an on-robot agent and a cloud service with content-addressed object storage with a PostgreSQL 14 index behind it. 6 items sit in the Config Distribution backlog; the entry that bears on the investment case is that a robot that is offline at publication applies the change without re-validating preconditions. Config Distribution reports 72 per cent line coverage, measured by the same instrumentation the pipeline uses. The assessment of Config Distribution is not unqualified: the reviewer accepts the engineering position while noting that the verifying test does not exist. Config Distribution was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Inputs to Config Distribution come from the site provisioning service and the artefact signing service, and its consumers are the on-robot supervisor and the motion planner parameters. The ageing position for Config Distribution is none; the bundle format was revised in 2025, and the reviewer checked it against the vendor statements on file. Platform Infrastructure holds Config Distribution, and the ownership entry naming Sam Oyelaran matches the on-call rotation. One coupling concern applies to Config Distribution: content-addressed object storage with a PostgreSQL 14 index is shared with a second service, so a schema change is not a local decision. Where Config Distribution is concerned, the commitment is that a configuration change reaches every online robot inside five minutes; the reviewer traced the metric to its source. Config Distribution was excluded from an earlier draft of TDR-2025-31 and reinstated after the reviewer obtained the supporting record.

Site Provisioning Service reports 64 per cent line coverage, measured by the same instrumentation the pipeline uses. Evidence examined for Site Provisioning Service comprised the provisioning runbook, template diffs for four sites and two go-live records. Responsibility for Site Provisioning Service was confirmed with Marcus Duplessis, whose team, Platform Infrastructure, also holds the on-call duty. One coupling concern applies to Site Provisioning Service: PostgreSQL 14 with a configuration history table is shared with a second service, so a schema change is not a local decision. Site Provisioning Service sits cloud, global, and no alternative placement is currently supported. Site Provisioning Service is included because a reader of TDR-2025-31 should be able to disagree with the inference without repeating the work.

On Site Provisioning Service the conclusion is qualified: management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. The dependency map places the identity service and the customer network intake records above Site Provisioning Service and the config distribution service and the WMS connector below it. Site Provisioning Service carries a single measurable objective, namely that a new site reaches a working configuration inside one working day. For Site Provisioning Service the supportability position is none identified. Site Provisioning Service carries 5 open defects, among them a case where site templates diverged after two

customers required bespoke network settings. Site Provisioning Service was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The assessment of Safety Controller is not unqualified: the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. The topology for Safety Controller is edge, dual-channel, with cross-checking between channels, and the reviewer confirmed it against the deployment manifests. The objective recorded against Safety Controller is that a validated protective-stop trigger de-energises the drive inside 120 ms, and it is reported weekly to the engineering leadership. Safety Controller depends on the E-stop chain, the speed and separation monitor and the drive current sensors and is depended on by the drive power contactors and the robot fault annunciator. Safety Controller's dependency ageing was checked: none; the supplier's support commitment runs to 2031. Safety Controller was raised again with the owning engineer after the initial interview, and the position did not change.

Safety Controller reports 91 per cent line coverage, measured by the same instrumentation the pipeline uses. Safety Engineering releases Safety Controller twice a year, with full revalidation of the safety case, and no release in the sampled window bypassed the gate set. Safety Controller is written in C, written to a constrained coding subset, executes on a safety-rated controller module supplied by a third party, and keeps its state in a non-volatile fault log on the controller. Safety Controller was assessed on the supplier assessment report, the internal safety case and eight bench trigger tests, and nothing beyond that was available in the data room. 3 defects remain open against Safety Controller; the one the reviewer pressed on is that the fault log wraps after four thousand entries and the oldest records are discarded. Safety Controller was reviewed against the position twelve months earlier, and nothing measurable has changed.

Changes to Analytics Warehouse Loader reach production weekly. Data and Telemetry tracks 20 open defects on Analytics Warehouse Loader, including the acknowledged case where a late-arriving telemetry partition is silently excluded rather than reprocessed. Data and Telemetry publishes one objective for Analytics Warehouse Loader: the daily load completes before 06:00 local time at every region. Analytics Warehouse Loader sits between the telemetry ingest path, the order broker and the billing usage meter and the customer reporting views and the internal metrics layer, and that position is what makes its availability objective load-bearing. Analytics Warehouse Loader sits at 45 per cent line coverage; the trend across the reviewed period was up by seven points. Analytics Warehouse Loader is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The evidence base for Analytics Warehouse Loader was the load schedule, partition completeness checks and two months of load audit records. The topology for Analytics Warehouse Loader is cloud, one loader per region writing to a shared warehouse, and the reviewer confirmed it against the deployment manifests. Behind Analytics Warehouse Loader sits a columnar warehouse with daily partitions; the service itself is Python 3.11 with SQL transformations on cloud, a scheduled batch runner. The Data and Telemetry team carries Analytics Warehouse Loader on its service inventory, and Farida Cisse answered for it in interview. Analytics Warehouse Loader publishes a stable interface, though two fields used by the customer reporting views and the internal metrics layer are documented only in code comments. Analytics Warehouse Loader was tested rather than accepted, and the test is described in the working papers behind TDR-2025-31.

Schedule C - Producer and consumer interface contracts examined (TDR-2025-31)

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Motion Planner	traffic manager	internal protocol	v2.8	enforced in the pipeline	none
Motion Planner	on-robot drive supervisor	request-response interface	v2.7	enforced in the pipeline	1 in the reviewed period
Costmap Fusion	motion planner	published schema	v3.9	not automated	none
Costmap Fusion	speed	streamed topic	v2.10	enforced in the pipeline	1 in the reviewed period
Fleet Allocator	traffic manager	request-response interface	v4.10	not automated	1 in the reviewed period
Fleet Allocator	fleet health monitor	shared event contract	v1.7	enforced in the pipeline	1 in the reviewed period
Traffic Manager	motion planner	shared event contract	v3.4	advisory only	2 in the reviewed period
Traffic Manager	remote operations console	streamed topic	v3.4	enforced in the pipeline	1 in the reviewed period
Charge Scheduler	fleet allocator	published schema	v2.6	not automated	1 in the reviewed period
Charge Scheduler	fleet health monitor	request-response	v1.8	not automated	1 in the reviewed

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
		interface			period
Safety Controller	drive power contactors	published schema	v2.5	advisory only	1 in the reviewed period
Safety Controller	robot fault annunciator	streamed topic	v1.2	enforced on the producer side	2 in the reviewed period
E-Stop Chain	safety controller	internal protocol	v4.9	not automated	1 in the reviewed period
E-Stop Chain	drive contactors	shared event contract	v4.5	not automated	1 in the reviewed period
Lidar Driver	costmap fusion pipeline	streamed topic	v4.8	advisory only	none
Lidar Driver	localization engine	published schema	v3.1	advisory only	1 in the reviewed period
Localization Engine	motion planner	streamed topic	v2.0	enforced on the producer side	1 in the reviewed period
Localization Engine	fleet health monitor	published schema	v2.7	not automated	2 in the reviewed period
WMS Connector	order broker	streamed topic	v4.1	not automated	1 in the reviewed period
WMS Connector	billing usage meter	shared event contract	v1.2	not automated	none
Order Broker	fleet allocator	request-response interface	v1.4	enforced in the pipeline	2 in the reviewed period
Order Broker	analytics warehouse loader	published schema	v4.11	enforced on the producer side	2 in the reviewed period
Telemetry Ingest	fleet health monitor	internal protocol	v3.8	enforced in the pipeline	none
Telemetry Ingest	alarm router	request-response interface	v2.7	enforced in the pipeline	none
OTA Updater	on-robot supervisor	shared event contract	v1.7	not automated	2 in the reviewed period
OTA Updater	fleet health monitor	streamed topic	v2.10	enforced on the producer side	none
Identity and Authorization	every cloud service	shared event contract	v4.10	enforced on the producer side	2 in the reviewed period
Identity and Authorization	remote operations console	published schema	v1.8	advisory only	2 in the reviewed period
Remote Operations Console	traffic manager	request-response interface	v2.11	not automated	none
Remote Operations Console	alarm router	published schema	v3.0	enforced on the producer side	1 in the reviewed period
Simulation Harness	release orchestrator gate	request-response interface	v4.10	enforced on the producer side	1 in the reviewed period
Simulation Harness	autonomy regression report	request-response interface	v2.4	not automated	none
Hardware-in-the-Loop Rig	firmware release decision record	published schema	v2.3	enforced on the producer side	2 in the reviewed period

4. Interface boundaries and data ownership around Battery Management Service - part 2 of 2

Below, the 4. Interface boundaries and data ownership around Battery Management Service continues; nothing separates this part from the last except the page on which it happens to fall. Nothing is restated from the earlier part.

Column definitions stated with the first part of the 4. Interface boundaries and data ownership around Battery Management Service govern the whole of it, and any qualification recorded there is carried forward here. Pagination alone determines where it falls.

Schedule C - Producer and consumer interface contracts examined (TDR-2025-31) (continued, part 2 of 2)

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Fleet Health Monitor	alarm router	streamed topic	v4.10	enforced on the producer side	1 in the reviewed period
Fleet Health Monitor	field service workflow	published schema	v4.6	advisory only	2 in the reviewed period
Battery Management Service	charge scheduler	request-response interface	v4.11	not automated	2 in the reviewed period
Battery Management Service	fleet health monitor	streamed topic	v2.1	enforced on the producer side	none
Gripper Controller	order execution path	published schema	v2.4	enforced in the pipeline	1 in the reviewed period

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Gripper Controller	safety controller fault input	published schema	v3.5	enforced on the producer side	1 in the reviewed period
Map Authoring Studio	map distribution channel	internal protocol	v3.4	advisory only	2 in the reviewed period
Map Authoring Studio	localization engine	internal protocol	v1.8	not automated	2 in the reviewed period
Site Provisioning Service	config distribution service	published schema	v1.6	enforced on the producer side	none
Site Provisioning Service	WMS connector	internal protocol	v2.10	enforced in the pipeline	1 in the reviewed period
Event Bus	every consuming service	published schema	v1.5	advisory only	none
Config Distribution	on-robot supervisor	streamed topic	v2.1	enforced on the producer side	2 in the reviewed period
Config Distribution	motion planner parameters	internal protocol	v1.3	enforced on the producer side	2 in the reviewed period
Robot Shadow Store	fleet allocator	shared event contract	v4.9	enforced on the producer side	none
Robot Shadow Store	remote operations console	streamed topic	v1.9	enforced on the producer side	2 in the reviewed period
Analytics Warehouse Loader	customer reporting views	streamed topic	v4.2	enforced in the pipeline	1 in the reviewed period
Analytics Warehouse Loader	internal metrics layer	shared event contract	v2.1	enforced on the producer side	none
Billing Usage Meter	finance export	streamed topic	v2.7	enforced in the pipeline	1 in the reviewed period
Billing Usage Meter	customer usage report	streamed topic	v3.8	advisory only	2 in the reviewed period
Alarm Router	paging provider	streamed topic	v2.3	not automated	1 in the reviewed period
Alarm Router	customer notification channel	request-response interface	v3.6	not automated	1 in the reviewed period
Video Snippet Service	remote operations console	shared event contract	v2.5	enforced in the pipeline	2 in the reviewed period
Video Snippet Service	incident review workflow	shared event contract	v3.9	enforced in the pipeline	none
Speed and Separation Monitor	safety controller	streamed topic	v1.3	enforced in the pipeline	none
Speed and Separation Monitor	motion planner speed limit	published schema	v1.6	advisory only	none
Wheel Odometry Filter	localization engine	request-response interface	v1.2	not automated	1 in the reviewed period
Wheel Odometry Filter	motion planner	published schema	v2.7	enforced on the producer side	1 in the reviewed period
Payload Scale Interface	gripper controller	request-response interface	v2.0	enforced on the producer side	1 in the reviewed period
Payload Scale Interface	order execution path	shared event contract	v4.9	not automated	none
Edge Gateway Agent	telemetry ingest path	streamed topic	v4.3	enforced in the pipeline	2 in the reviewed period
Edge Gateway Agent	robot shadow store	streamed topic	v2.1	advisory only	none
Release Orchestrator	OTA updater	request-response interface	v1.4	enforced in the pipeline	2 in the reviewed period
Release Orchestrator	cloud deployment controllers	internal protocol	v4.0	advisory only	none

5. Automated test coverage across the reviewed services

Where Robot Shadow Store is concerned, the commitment is that shadow state is no more than two seconds behind the robot at p95; the reviewer traced the metric to its source. One coupling concern applies to Robot Shadow Store: Redis 7 as the hot path with PostgreSQL 14 as the durable record is shared with a second service, so a schema change is not a local decision. Fleet Services releases Robot Shadow Store weekly, and no release in the sampled window bypassed the gate set. Robot Shadow Store reports 68 per cent line coverage, measured by the same instrumentation the pipeline uses. Robot Shadow Store was assessed on the shadow consistency note, staleness measurements and a rebalance observation, and nothing beyond that was available in the data room. Robot Shadow Store is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Localization Engine does not share a schema with any consumer; every read by the motion planner and the fleet health monitor goes through the published interface. Localization Engine was assessed on the localization design note, survey reports from three sites and the reviewer's map audit, and nothing beyond that was available in the data room. Localization Engine ships every three weeks with the planner; the reviewer sampled the last twelve releases. 16 defects remain open against Localization Engine; the one the reviewer pressed on is that relocalisation after a lift-and-place takes longer than the operator prompt allows. The manifests place Localization Engine edge, with map distribution from the cloud. Localization Engine is reported here in full because a summary of it in an earlier draft of TDR-2025-31 was misread as an assurance.

The site survey imports and the identity service feed Map Authoring Studio; a failure in Map Authoring Studio is felt by the map distribution channel and the localization engine. For Map Authoring Studio the stated target is that a published map revision reaches every robot at the site inside fifteen minutes, expressed against production traffic rather than a benchmark. Placement of Map Authoring Studio is cloud, single global deployment, which the reviewer tested by tracing a request end to end. For Map Authoring Studio the reviewer records that management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. Map Authoring Studio publishes a stable interface, though two fields used by the map distribution channel and the localization engine are documented only in code comments. Map Authoring Studio was compared with the equivalent position at two other components, and it is neither the best nor the worst.

The ageing position for Simulation Harness is the physics core depends on a library branch that stopped receiving fixes in 2023, and the reviewer checked it against the vendor statements on file. Hosting for Simulation Harness is developer workstations and a batch cluster, and its Python 3.11 with a C++ physics core code writes through to object storage for scenario definitions and run artefacts. Simulation Harness depends on the planner and localization builds under test and is depended on by the release orchestrator gate and the autonomy regression report. The published objective for Simulation Harness is that the nightly scenario suite completes inside four hours. Simulation Harness is promoted continuous, from the main branch, with the ring order fixed by policy. Simulation Harness is the reason one section of TDR-2025-31 concludes narrowly rather than generally.

Schedule D - Automated coverage trend on the reviewed services (TDR-2025-31)

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Motion Planner	63%	70%	71%	+8	Autonomy Core
Costmap Fusion	51%	55%	58%	+7	Perception
Fleet Allocator	74%	79%	82%	+8	Fleet Services
Traffic Manager	65%	72%	74%	+9	Fleet Services
Charge Scheduler	54%	60%	63%	+9	Fleet Services
Safety Controller	82%	88%	91%	+9	Safety Engineering
E-Stop Chain	n/a	n/a	witnessed test	not comparable	Safety Engineering
Lidar Driver	37%	44%	44%	+7	Robot Firmware
Localization Engine	61%	66%	66%	+5	Autonomy Core
WMS Connector	60%	68%	69%	+9	Integrations
Order Broker	72%	75%	78%	+6	Fleet Services
Telemetry Ingest	55%	58%	61%	+6	Data and Telemetry
OTA Updater	67%	73%	76%	+9	Platform Infrastructure
Identity and Authorization	77%	82%	84%	+7	Security Engineering
Remote Operations Console	48%	51%	52%	+4	Applications
Simulation Harness	44%	46%	47%	+3	Developer Productivity
Hardware-in-the-Loop Rig	35%	36%	39%	+4	Robot Firmware
Fleet Health Monitor	50%	54%	57%	+7	Field Reliability
Battery Management Service	45%	51%	54%	+9	Robot Firmware
Gripper Controller	38%	38%	41%	+3	Robot Firmware
Map Authoring Studio	43%	48%	49%	+6	Applications
Site Provisioning Service	55%	62%	64%	+9	Platform Infrastructure
Event Bus	n/a	n/a	witnessed test	not comparable	Platform

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
					Infrastructure
Config Distribution	68%	69%	72%	+4	Platform Infrastructure
Robot Shadow Store	63%	68%	68%	+5	Fleet Services
Analytics Warehouse Loader	42%	43%	45%	+3	Data and Telemetry
Billing Usage Meter	75%	77%	79%	+4	Data and Telemetry
Alarm Router	46%	50%	51%	+5	Field Reliability
Video Snippet Service	37%	43%	43%	+6	Applications
Speed and Separation Monitor	82%	87%	87%	+5	Safety Engineering
Wheel Odometry Filter	53%	60%	62%	+9	Robot Firmware
Payload Scale Interface	29%	33%	36%	+7	Robot Firmware
Edge Gateway Agent	67%	73%	74%	+7	Platform Infrastructure
Release Orchestrator	47%	53%	55%	+8	Developer Productivity

6. What the incident record shows about the architecture

Internal felt Video Snippet Service incident INC-2025-152, which affected no production site and no robot directly, the exposure being to internal systems. Assessed on its own record, incident INC-2025-152 shows that the failure was in the detection design rather than in the code that failed. Video Snippet Service owns incident INC-2025-152, which the reviewer classes as data retention. The reviewer checked the incident INC-2025-152 commitment against the code and release records; the rule landed on 2026-01-10; deletion of the historical clips is open. Duration for data retention event INC-2025-152 splits into not applicable; retention only of impact and 37 days to close the underlying cause. internal incident INC-2025-152 illustrates why TDR-2025-31 separates the severity of a finding from the confidence behind it.

In data retention event INC-2025-152 the first signal was an internal privacy review found the discrepancy, and the reviewer treats that as reported detection. 2025-12-04 is the date of INC-2025-152, on which one region retained operator video clips for ninety days against a thirty-day commitment. In data retention incident INC-2025-152, bucket creation was not covered by the configuration baseline check, which is a process finding rather than a code finding. The post-mortem for internal incident INC-2025-152 identifies the cause as follows: the lifecycle rule was applied per storage bucket and one region's bucket was created later without it. Evidence strength on the INC-2025-152 analysis is strong on the cause and limited on the contributing factor. INC-2025-152 has no single accountable owner, and that is the finding the reviewer would press hardest.

Duration for availability incident INC-2025-071 splits into 52 m of impact and 52 m to close the underlying cause. The reviewer grades the availability incident INC-2025-071 evidence as limited; the log retention window had closed before the reviewer asked for the raw data and has not upgraded the finding to compensate. Availability event INC-2025-071 records that on 2025-06-11, a regional ingest endpoint failed over and edge buffers filled during a fifty-two minute outage. Availability event INC-2025-071 would have been shorter or smaller had it not been that the upgrade runbook did not require a partition leadership check between broker restarts. Alerted detection applies to incident INC-2025-071, since the ingest availability alert fired within ninety seconds. INC-2025-071 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-31.

Availability incident INC-2025-071 affected 6 sites and 214 robots. Erik Palmgren committed after Telemetry Ingest incident INC-2025-071 to add a leadership and under-replicated partition check to the broker upgrade runbook, due 2025-07-31. Availability incident INC-2025-071 was caused by a design gap: a broker partition lost its leader during a rolling upgrade of the streaming cluster. The reviewer's reading of availability event INC-2025-071 is that the failure was in the detection design rather than in the code that failed. Verification of the availability event INC-2025-071 remediation gives landed on 2025-07-18 and exercised in the September failover drill. availability event INC-2025-071 is the clearest instance in TDR-2025-31 of a control that exists on paper and is weaker in practice.

Robot Shadow Store incident INC-2025-124 is not read as an isolated failure: the remediation addresses the immediate cause and leaves the contributing factor in place. How consistency incident INC-2025-124 surfaced matters to

the assessment: the console latency alert fired, though the underlying cause took a day to attribute. Following incident INC-2025-124 the team undertook to annotate staleness on every read path and add a rebalance-under-load test to the platform suite, with Marta Kellerman accountable by 2025-11-30. For consistency incident INC-2025-124 the follow-through was tested rather than assumed, and landed on 2025-10-30; the load test was added on 2025-11-12. Consistency incident INC-2025-124 records that on 2025-10-01, a shard rebalance served shadow state up to ninety seconds stale without marking it stale. consistency event INC-2025-124 is treated as material to the investment case rather than as a routine engineering item.

A contributing factor in multiple sites incident INC-2025-124 was that no test exercised a shard rebalance while the cluster was under load. Multiple sites felt Robot Shadow Store incident INC-2025-124, which affected 5 sites and 178 robots. Consistency incident INC-2025-124 carries moderate, resting on the team's own post-mortem and one supporting log extract evidence, and the reviewer declined to raise the finding beyond what that supports. Duration for multiple sites incident INC-2025-124 splits into 1 h 08 m of impact and 29 days to close the underlying cause. The classification applied to consistency event INC-2025-124 is consistency, and the affected component is Robot Shadow Store. INC-2025-124 is treated as material to the investment case rather than as a routine engineering item.

Observability is the class of observability event INC-2026-028; Fleet Health Monitor is the component. On Fleet Health Monitor incident INC-2026-028 the evidence is moderate, resting on the team's own post-mortem and one supporting log extract; the severity assessment is stated separately below. After Fleet Health Monitor incident INC-2026-028, Wanda Njoku took the action to group on the signal and block closure while a constituent alert remains active by 2026-04-30. The trigger recorded for INC-2026-028 on 2026-03-10 was that two distinct faults at one site were grouped into a single case and the second went unworked for two days. The detection path for Fleet Health Monitor incident INC-2026-028 was the customer asked why one symptom persisted after the case had been closed. observability event INC-2026-028 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Mitigation of Fleet Health Monitor incident INC-2026-028 took not a service outage; 2 days of delayed repair and full resolution took 31 days. Landed on 2026-04-10 is the position on the observability incident INC-2026-028 commitment as at the review date. The engineering team's root cause for incident INC-2026-028 is that alert grouping keyed on site and severity rather than on the underlying signal, and the reviewer found the analysis sound. INC-2026-028 affected one site and 22 robots. Taken with the rest of the record, observability event INC-2026-028 indicates that the team's analysis was honest about a gap it could have concealed. incident INC-2026-028 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Following Evergreen Warehousing incident INC-2026-045 the team undertook to aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current hardware, with Sam Oyelaran accountable by 2026-06-30. The reviewer's reading of Edge Gateway Agent incident INC-2026-045 is that the customer-reported detection path is the finding, not the defect itself. Beyond the direct cause, Edge Gateway Agent incident INC-2026-045 shows that buffer pressure was never aggregated to a site view, so the condition was not observable. Incident INC-2026-045 touched one site and 39 robots, and Evergreen Warehousing was the affected account. Edge Gateway Agent owns incident INC-2026-045, which the reviewer classes as observability. observability event INC-2026-045 was checked against the company's own register, which had not been updated since the change landed.

The alert landed on 2026-05-17; the buffer extension is in progress is the position on the Edge Gateway Agent incident INC-2026-045 commitment as at the review date. Evergreen Warehousing incident INC-2026-045 carries limited, being a single account corroborated only by the ticket timeline evidence, and the reviewer declined to raise the finding beyond what that supports. Underneath observability incident INC-2026-045 lies the design decision that the buffer is sized for four hours and drops oldest-first without raising an alert. The detection path for observability incident INC-2026-045 was the buffer pressure metric was visible per robot but no site-level alert existed. Observability event INC-2026-045 shows a short mitigation and a long tail: 5 h 12 m against 26 days. Evergreen Warehousing incident INC-2026-045 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Simulation Harness incident INC-2026-021 affected no production site and no robot directly, the exposure being to internal systems. Not landed; the determinism check is written but not enabled is the position on the internal incident INC-2026-021 commitment as at the review date. The commitment register shows INC-2026-021 against an undertaking to pin the full build configuration and add a determinism check to the harness, with Colin Achebe named and 2026-05-31

committed. On 2026-02-24 the record for Simulation Harness incident INC-2026-021 opens: a nightly simulation suite produced non-reproducible results after the physics core was rebuilt. A contributing factor in INC-2026-021 was that the build pinned the compiler but not the optimisation flags used for release builds. tooling event INC-2026-021 is treated as material to the investment case rather than as a routine engineering item.

Underneath incident INC-2026-021 lies the design decision that the physics core is not deterministic across optimisation levels. Simulation Harness incident INC-2026-021 supports one conclusion and not another: an operational workaround masked the defect for longer than the incident record suggests. Simulation Harness incident INC-2026-021 reached the engineering team when the autonomy team noticed a step change in the pass rate. The two durations recorded for Simulation Harness incident INC-2026-021 are not applicable; developer tooling to mitigate and open at the review date to resolve. Simulation Harness accounts for incident INC-2026-021, a tooling event. incident INC-2026-021 carries a follow-up request that was satisfied after the draft of TDR-2025-31 was circulated.

What went wrong in incident INC-2026-009 was that force limiting depends on a calibration that is only triggered by a service workflow step. Blue Mesa Components incident INC-2026-009 records that on 2026-01-27, a gripper applied more force than the configured limit after a mechanical replacement without recalibration. Blue Mesa Components incident INC-2026-009 was contained to one site and 4 robots. Ines Duarte committed after calibration incident INC-2026-009 to require a calibration record before returning a manipulator robot to service, due 2026-04-15. Gripper Controller owns calibration incident INC-2026-009, which the reviewer classes as calibration. Blue Mesa Components incident INC-2026-009 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

A contributing factor in Gripper Controller incident INC-2026-009 was that the service workflow allowed the step to be marked complete without a calibration record. Grading Gripper Controller incident INC-2026-009 on evidence rather than on consequence gives strong, resting on the production traces the team supplied. Gripper Controller incident INC-2026-009 was detected because the payload damage was reported by the customer. Landed on 2026-03-16 with a workflow gate is the position on the Gripper Controller incident INC-2026-009 commitment as at the review date. The reviewer's reading of Blue Mesa Components incident INC-2026-009 is that the failure was in the detection design rather than in the code that failed. Gripper Controller incident INC-2026-009 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

7. Third-party components inside the reviewed services

The distribution question for lidar-vendor-sdk turns on the fact that it is used lidar driver. The licence on lidar-vendor-sdk is commercial, redistributable binary; copyleft exposure is no. Lidar-vendor-sdk entered the codebase by this route: supplied by the sensor vendor. lidar-vendor-sdk was reviewed against the position twelve months earlier, and nothing measurable has changed.

Wms-adapter-sdk carries commercial, per-seat terms, classified no for copyleft purposes. In WMS Connector the component wms-adapter-sdk is used one customer's warehouse system adapter, which is the fact the licence analysis rests on. OSS-017 records wms-adapter-sdk 2.2.0 against WMS Connector, one customer's warehouse system adapter. wms-adapter-sdk was checked against the corresponding entry in the company's own register, and the register was current.

Usage of geom2d-fusion 0.9.7 is dynamically linked by the fusion pipeline, and the inventory reference is OSS-002. Beyond licensing, geom2d-fusion carries a single-maintainer dependency with no succession plan. Compliance with the geom2d-fusion terms requires to share alike any derived asset it distributes. geom2d-fusion would be a candidate for the first post-close engineering review under any reasonable plan.

Meeting the terms on rtos-shim means to attribute the component in the customer-facing notice file. OSS-008 covers rtos-shim at version 1.0.9, used real-time helper layer on the robot. Rtos-shim was obtained as follows: public registry, signature verified. rtos-shim would move from a concern to a verified defect if one further measurement were supplied.

Beyond licensing, protobuf-runtime carries a fork that must be maintained locally for as long as it is used. Meeting the terms on protobuf-runtime means to reproduce the notice text in the delivered documentation. The distribution question for protobuf-runtime turns on the fact that it is used message encoding across edge and cloud. protobuf-runtime has no single accountable owner, and that is the finding the reviewer would press hardest.

Licence text and the actual usage of safety-controller-toolchain were read together, and development-use only; no runtime redistribution occurs. Commercial, development use only governs safety-controller-toolchain, which the inventory marks as no copyleft. No identified risk beyond ordinary version currency attaches to safety-controller-toolchain independently of its licence terms. safety-controller-toolchain is reported at the evidence strength available and has not been promoted to fit the narrative.

Meeting the terms on pager-provider-sdk means to keep the dynamic-linking boundary intact. No copyleft applies to pager-provider-sdk, whose licence is commercial, subscription. Against pager-provider-sdk the reviewer would add that the older application interface is deprecated by the provider with an end date in 2027. pager-provider-sdk sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Approval for cell-model-lib reads approved 2024-10-01, and the reviewer checked the approving record. Cell-model-lib obliges the company to keep the dynamic-linking boundary intact. Usage of cell-model-lib 1.8.0 is battery state estimation, and the inventory reference is OSS-020. cell-model-lib would move from a concern to a verified defect if one further measurement were supplied.

8. Architectural debt and the cost of removing it

TD-16 on Localization Engine is the item to encrypt the on-robot localization map cache, classified under security. 1 engineer-quarter is the estimate on TD-16, which converts to $126,000. On funding, register item TD-16 is partly funded, with the second phase unbudgeted. security item TD-16 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The estimate for IP and support item TD-02 is 1.5 engineer-quarters and $185,000. The reviewer's qualification on register item TD-02 is that the reviewer treats this as both an IP and a supportability item. Sequencing of Localization Engine item TD-02 matters: it is in parallel with the release-process changes. register item TD-02 was checked against the company's own register, which had not been updated since the change landed.

Security item TD-15 is on the register because it one credential compromise affects paging everywhere. Whether security item TD-15 happens depends on funding, and it is unfunded and dependent on capital rather than engineering effort. In the programme TD-15 on Alarm Router is last in a three-item chain, so its date is the least reliable on the register. security item TD-15 would be a candidate for the first post-close engineering review under any reasonable plan.

Register item TD-09 is committed with a target date of 2027-06-30. Sizing on Event Bus item TD-09 gives 1.5 engineer-quarters and $196,000. Register item TD-09 is funded and already staffed; the register status and the funding position are not the same thing. TD-09 on Event Bus was raised again with the owning engineer after the initial interview, and the position did not change.

TD-13 on Localization Engine cannot start earlier than it does because it is behind a customer maintenance window that is not yet agreed. Effort on TD-13 is estimated at 1.5 engineer-quarters, costed at $178,000. TD-13 appears in the register and is funded and already staffed. Localization Engine item TD-13 was raised again with the owning engineer after the initial interview, and the position did not change.

About Hardware-in-the-Loop Rig item TD-05 it should be said that outstanding since the April 2025 lidar timing incident. TD-05 on Hardware-in-the-Loop Rig is the item to bring the second bench rig to the current sensor configuration, classified under test capability. Partly funded, with the second phase unbudgeted is the funding position on register item TD-05. test capability item TD-05 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

9. Where the architecture meets the safety case

Protection at register entry HZ-20 comes from return-intensity monitoring with a cleaning prompt and a degraded-mode speed limit. One qualification on Occasional-exposure hazard HZ-20 the reviewer would not remove is that cleaning compliance records were incomplete at one site. The residual entry for hazard HZ-20 in the register reads Tolerable where cleaning compliance is evidenced, partially verified. HZ-20 was checked against the corresponding entry in the company's own register, and the register was current.

The register places at hazard HZ-16 the scenario in which firmware update leaves a robot in a mixed-version state. Produced two near-miss reports from site staff is the incident history behind Equipment damage hazard HZ-16. One qualification on Remote-exposure hazard HZ-16 the reviewer would not remove is that the ring pause defect in July 2025 did not produce a mixed-version robot. Equipment damage hazard HZ-16 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Hazard HZ-09 in the register carries residual risk of Not yet tolerable without the slip detector with a status of open. The record shows that hazard HZ-09 in the register has produced no event, though the control has activated repeatedly. Reading HZ-09 against the verification evidence, the slip detector is prototyped and unvalidated as at the review date. register entry HZ-09 was checked against the corresponding entry in the company's own register, and the register was current.

The residual entry for HZ-12 reads Tolerable, verified. What the register does not say about Minor injury hazard HZ-12, and the reviewer does, is that one site's dock layout leaves less clearance than the layout guide requires. Risk parameters for Occasional-exposure hazard HZ-12 are Minor injury severity, Occasional exposure and Low controllability. Minor injury hazard HZ-12 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The hazard recorded at Occasional-exposure hazard HZ-22 is that site network loss leaves robots operating without fleet coordination. The parameter set for hazard HZ-22 in the register reads Equipment damage, Occasional, Medium. Equipment damage hazard HZ-22 is relevant to the newest three sites, where commissioning is incomplete rather than to the whole fleet. Occasional-exposure hazard HZ-22 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Fatality harm, Improbable exposure and Low controllability are the parameters recorded for Improbable-exposure hazard HZ-05. Improbable-exposure hazard HZ-05 has produced no event, though the control has activated repeatedly; that is worth stating alongside the assessed exposure. The control applied at register entry HZ-05 is hardwired dual-channel stop chain independent of the compute module. hazard HZ-05 in the register would move from a concern to a verified defect if one further measurement were supplied.

Risk parameters for register entry HZ-18 are Serious injury severity, Occasional exposure and Low controllability. Where HZ-18 bites is the two pallet-mover variants only. In the reviewed period register entry HZ-18 has never activated in the field and is evidenced only on the bench. Occasional-exposure hazard HZ-18 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

On history, hazard HZ-06 in the register produced a single event before the current control was fitted. The residual entry for hazard HZ-06 reads Tolerable, verified. The six sites with narrow-aisle geometry is the exposed population for register entry HZ-06. Serious injury hazard HZ-06 illustrates why TDR-2025-31 separates the severity of a finding from the confidence behind it.

10. Open items, declined remediations and qualifications

The open items carried forward from report TDR-2025-31 are set out below; a reader should treat the committed dates as intentions rather than facts. This report preserves 8 unresolved questions rather than resolving them by assumption. This report, this report, is the Pinehurst Systems Assurance workstream on the service architecture, ownership boundaries and dependency structure of the fleet platform, delivered to Ardent Peak on 2025-12-17. Pinehurst Systems Assurance was instructed by Ardent Peak to examine the service architecture, ownership boundaries and dependency structure of the fleet platform; the resulting report is the present report, dated 2025-12-17. The company reviewed a draft of TDR-2025-31 for factual accuracy and its corrections are reflected without altering any conclusion.

TDR-2025-31 treats a backlog label as evidence of intent and not as evidence of remediation, which is why several items remain open in this report that the company regards as handled. Two positions in report TDR-2025-31 are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. L. Casaubon tested a sample rather than the population in the report at TDR-2025-31, and the sampling basis is stated in each section. The report at TDR-2025-31 did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. L. Casaubon has stated the limitations at the point they bite rather than collecting them into a disclaimer.

Reporting under the report at TDR-2025-31 should distinguish a target that was met from a target that was lowered, which the current pack does not. Post-close monitoring under report TDR-2025-31 would track four things: objective attainment, change failure rate, open finding ageing and coverage on the control path. The population for the report at TDR-2025-31 was the service inventory as at 2025-12-17; 19 entries were examined and the rest were reviewed only at inventory level. For the report at TDR-2025-31 the reviewer sampled 3 sites, including the largest deployment and the newest. L. Casaubon has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

Report TDR-2025-31 grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. L. Casaubon treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and this report keeps the two apart. Where this report reports a supplier's test result, the reviewer read the certificate and did not witness the test. The present report relies on the company's incident tracker for completeness of the incident population, which the reviewer could corroborate but not prove. Pinehurst Systems Assurance has reported the position as at 2025-12-17 and has not attempted to predict where it will be at completion.

11. Conclusions and implications for the transaction

Taking the sections above together, the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. For Ardent Peak's purposes, this report concludes that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. Selection for TDR-2025-31 favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. The report at TDR-2025-31 sampled 19 services from an inventory of 34, weighted towards those on the critical control path. Each section of the present report states its own basis, so a reader can accept one section and reject another.

6 items in this report are post-close engineering work rather than transaction issues, and the reviewer has costed them at $1,330,000. Report TDR-2025-31 suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. This report addresses the service architecture, ownership boundaries and dependency structure of the fleet platform. It does not address commercial terms, financial reporting or employment matters, which sit with other workstreams. Pinehurst Systems Assurance delivered the present report on 2025-12-17; the scope agreed with Ardent Peak was the service architecture, ownership boundaries and dependency structure of the fleet platform. Figures quoted in report TDR-2025-31 come from the company's systems unless the text says the reviewer measured them.

The pack report TDR-2025-31 proposes is deliberately short, because a longer one would be produced and not read. Monitoring recommended by this report relies on metrics the company already emits, so no new instrumentation is required to start. Open items in this report are stated with what would close each one, so a post-close plan can be costed against them. Pinehurst Systems Assurance will not treat an open item in the present report as closed on a verbal assurance, and has said so to the engineering leadership. Pinehurst Systems Assurance has reported unresolved disagreements as disagreements rather than choosing between the accounts.